

Day 4 — Turmeric — Golden Root

Module: Herbs — Auṣadhi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: distinguish fresh root vs powder; identify curcumin as active compound.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *haridrā* (IAST: haridrā; lit. “turmeric — *Curcuma longa*”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Botanical identification.** *Curcuma longa*. A rhizome (underground stem, not a root). Bright orange when cut. Fresh vs dried (turmeric powder).
2. **Active compound.** Curcumin (~2-5% of dried rhizome). Bright yellow polyphenol. Other compounds: turmerones in the essential oil.
3. **Traditional uses.** Caraka cites *haridrā* for: wound healing (topical paste), anti-inflammatory (golden milk), as a colouring + preservative agent in food. Centuries of culinary use.
4. **Modern research.** Curcumin has been studied extensively for anti-inflammatory effects. Bioavailability is poor without piperine (black pepper) or fat — which is why *haldi doodh* (turmeric milk with ghee) is traditional. Modern preparations use liposomal or piperine-enhanced formulations.
5. **Demo / activity.** Crush a small piece of fresh turmeric on paper. Note the staining (curcumin is a pH indicator — colour changes with vinegar/lemon vs baking soda). This is also chemistry, not just culinary.

Activity (15 min)

A scaled version of *Blind Smell-Test Identification* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Turmeric — Golden Root.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).